

Achieving both High Ionic Conductivity and High Interfacial Stability with the $\text{Li}_{2+x}\text{C}_{1-x}\text{B}_x\text{O}_3$ Solid-State Electrolyte: Design from Theoretical Calculations

Bingkai Zhang, Zhan Lin, Lin-Wang Wang, and Feng Pan

ACS Appl. Mater. Interfaces, **Just Accepted Manuscript** • DOI: 10.1021/acsami.9b22185 • Publication Date (Web): 15 Jan 2020

Downloaded from pubs.acs.org on January 16, 2020

Just Accepted

“Just Accepted” manuscripts have been peer-reviewed and accepted for publication. They are posted online prior to technical editing, formatting for publication and author proofing. The American Chemical Society provides “Just Accepted” as a service to the research community to expedite the dissemination of scientific material as soon as possible after acceptance. “Just Accepted” manuscripts appear in full in PDF format accompanied by an HTML abstract. “Just Accepted” manuscripts have been fully peer reviewed, but should not be considered the official version of record. They are citable by the Digital Object Identifier (DOI®). “Just Accepted” is an optional service offered to authors. Therefore, the “Just Accepted” Web site may not include all articles that will be published in the journal. After a manuscript is technically edited and formatted, it will be removed from the “Just Accepted” Web site and published as an ASAP article. Note that technical editing may introduce minor changes to the manuscript text and/or graphics which could affect content, and all legal disclaimers and ethical guidelines that apply to the journal pertain. ACS cannot be held responsible for errors or consequences arising from the use of information contained in these “Just Accepted” manuscripts.

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

Achieving both High Ionic Conductivity and High Interfacial Stability with the $\text{Li}_{2+x}\text{C}_{1-x}\text{B}_x\text{O}_3$ Solid-State Electrolyte: Design from Theoretical Calculations

Bingkai Zhang,^{*,†,‡} Zhan Lin,^{*,†} Lin-Wang Wang,^{\$} and Feng Pan^{*,‡}

[†]School of Chemical Engineering and Light Industry, Guangdong University of Technology, Guangzhou 510006, China

[‡]School of Adv. Mater., Peking University Shenzhen Graduate School, Shenzhen 518055, China

^{\$}Joint Center for Artificial Photosynthesis and Materials Sciences Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720, United States

KEYWORDS: $\text{Li}_{2+x}\text{C}_{1-x}\text{B}_x\text{O}_3$, Solid electrolyte, High interfacial stability, Separator, Solid-state lithium battery

ABSTRACT

Crystalline solid electrolyte interphase Li_2CO_3 material with a large bandgap shows promise towards next-generation all-solid-state lithium batteries (ASSLBs). However, the inferior ionic diffusivity restricts such structure into a real battery setup. Herein, based on density functional theory (DFT)

calculation and Python materials genomics (pymatgen), we theoretically develop the chemistry and local structural motifs to build a mixed boron-carbon framework $\text{Li}_{2+x}\text{C}_{1-x}\text{B}_x\text{O}_3$ (LCBO). We examine the electrochemical and chemical stability of LCBO-electrode interfaces by analyzing the thermodynamics of formation of interfacial phases. Interestingly, the LCBO material is automatically protected from further decomposition through the self-generated resistive interphase (Li_2CO_3 and Li_3BO_3), which gives a widening range of operating potential. The LCBO shows high interfacial stability with LiCoO_2 , LiMnO_2 , and LiMn_2O_4 . More importantly, the theoretical Li-ion migration barrier of LCBO ($x = 0.375$) is about 0.23 ± 0.02 eV through a cooperative migration mechanism. Therefore, LCBO material combines high Li-ion diffusivity along with good interfacial stability, which makes it a promising solid-state electrolyte material for ASSLBs.

1. INTRODUCTION

ASSLBs as potential candidates have recently been extensively studied due to their high lithium-ion (Li-ion) capacity and high safety.¹⁻³ The most important things in ASSLBs are the inorganic solid electrolytes (ISEs) with high ionic conductivity and good stability.⁴⁻⁷ However, only a handful of crystalline structure families, such as LISICON-like,⁸⁻¹⁰ NASICON-like,¹¹⁻¹² argyrodite,¹³⁻¹⁵ garnet,¹⁶⁻¹⁷ and perovskite,¹⁸⁻¹⁹ have been reported as ISEs. Finding potential solid-state electrolytes with excellently interfacial compatibility is an important task for ASSLBs development.²⁰⁻²¹

Li_2CO_3 is widely believed to be solid electrolyte interphase (SEI) and is used as an artificial coating layer on electrodes due to its electrochemical stability.²²⁻²⁶ Based on the properties of Li_2CO_3 coating material, the design of crystalline Li_2CO_3 as ISEs in ASSLBs is an intriguing possibility. Other than

for coating material use, experimental studies show that Li-ion conductivity in crystalline Li_2CO_3 is $\sim 10^{-10} \text{ S cm}^{-1}$ (temperature = 298 K)²⁷ or $\sim 2 \times 10^{-7} \text{ S cm}^{-1}$ (temperature = 473 K)²⁸, which far behind that of bulk ISEs (10^{-4} to $10^{-2} \text{ S cm}^{-1}$ at room temperature).⁵ The low ionic conductivity of crystalline Li_2CO_3 is due to the low intrinsic point defects such as Li interstitials since Li_2CO_3 can hardly exhibit off-stoichiometric composition during battery. These insights motivate us to consider increasing the concentration of Li interstitials by replacing CO_3^{2-} with BO_3^{3-} , which is $\text{Li}_{2+x}\text{C}_{1-x}\text{B}_x\text{O}_3$ (LCBO), so that it will enhance Li conductivity. Recently, Jung et al.²⁹ synthesized $\text{Li}_{3-x}\text{B}_{1-x}\text{C}_x\text{O}_3$ (LBCO) based on the Li_3BO_3 compound and applied the Li_3BO_3 - Li_2CO_3 as coatings for LiCoO_2 cathode in ASSLBs, which drastically enhances the electrochemical performances. Yet, the detailed LCBO phase and interfacial structure and ion conductivity are little investigated so far in the view of ISEs.

Here, by applying first-principles density functional theory (DFT),³⁰⁻³⁴ *ab initio* molecular dynamics (AIMD),³⁵⁻³⁹ and pymatgen materials analysis library,⁴⁰⁻⁴⁴ we aim to probe the effect of BO_3^{3-} units within Li_2CO_3 framework on the phase stability, electrochemical stability, chemical stability in contact with electrodes, and ionic conductivity. These modeling and analyzing techniques are well suited to the investigation of such properties and have been applied successfully to a variety of studies on solid-state electrolytes.⁴⁵ Our results suggest that crystalline LCBO material inherits good interfacial stability with electrodes from Li_2CO_3 . LCBO material has a wide electrochemical window through the formation of resistive interphase (Li_2CO_3 and Li_3BO_3) at a reduction potential of 1.27 V. The LCBO also shows very high Li-ionic conductivity at room temperature. The good Li-ion diffusivity and high electrochemical stability make LCBO an effective solid-state electrolyte material for ASSLBs.

2. COMPUTATIONAL METHODS

All calculations this work were performed by Vienna *ab initio* simulation package (VASP).⁴⁶ The electron-ion interaction was described with the projector augmented wave (PAW).⁴⁷ Valence electrons were described by the plane-wave basis set and core electrons were kept frozen in the configuration for which the PAW dataset was generated. The electronic orbitals were expanded by plane waves with a maximum kinetic energy of 500 eV. The exchange-correlation part of the density functional was treated within the generalized gradient approximation (GGA) of Perdew-Burke-Ernzerhof (PBE).⁴⁸ The convergence was used for terminating the self-consistency interactions was below 10^{-6} eV/atom. For DFT simulations, a unit cell of 96 atoms with Monkhorst-Pack k-point meshes ($2 \times 2 \times 2$) was used. For AIMD simulation, a unit cell of 208 atoms with a Gamma-centered k-point mesh was used. To get more information on DFT method, the two references (Ref. 49 and 50) offer more information.⁴⁹⁻⁵⁰

To study the thermodynamic stability upon B-C mixing with respect to parent ternary oxide, we calculate the quantity ΔE_{mixing} which is defined as,

$$\Delta E_{\text{mixing}}(\text{LCBO}) = (E(\text{LCBO}) - (1-x) \cdot E(\text{Li}_2\text{CO}_3) - x \cdot E(\text{Li}_3\text{BO}_3))N, \quad (1)$$

where N is the total number of atoms in the system. The electrochemical stability of the LCBO framework is studied using the grand potential phase diagram⁴³⁻⁴⁴ and the thermochemical data in Open Quantum Materials Database (OQMD) as well as in python materials genomics (pymatgen).⁵¹⁻⁵³ The OQMD is a database of DFT-calculated thermodynamic and structural properties. We first identify the electrochemical window of LCBO framework in a Li chemical potential range, μ_{Li} , in

which the composition of LCBO is stable against either Li extraction or insertion. The μ_{Li} is determined according to the equation:

$$\mu_{\text{Li}}(\phi) = \mu^0_{\text{Li}} - e\phi \quad (2)$$

where μ^0_{Li} is the chemical potential of Li-metal, and potential ϕ is open to Li.

Then, we evaluate the chemical stability of ISE-electrode interfaces to estimate whether materials react exothermically with or without the applied chemical potential.⁵⁴⁻⁵⁶ Here, we consider the interface as a pseudo-binary of ISE and electrode, $x \cdot \text{ISE} + (1-x) \cdot \text{electrode}$. For LCBO-cathode interfaces at none applied potentials, the mutual reaction energy is defined in terms of $\Delta E_{\text{mutual energy}}$ in equation (7) after considering the reaction equilibrium as well as the following equations (3-6):

$$\Delta E_{\text{LCBO}} = E(\text{EP}) - E(\text{LCBO}), \quad (3)$$

$$\Delta E_{\text{cathode}} = E(\text{EP}) - E(\text{Ca}), \quad (4)$$

$$E_{\text{interface}}(\text{LCBO-Ca}, x) = x \cdot E(\text{LCBO}) + (1-x) \cdot E(\text{Ca}), \quad (5)$$

$$\Delta E_{\text{interface}}(\text{LCBO-Ca}, x) = E(\text{EP}, x) - E_{\text{interface}}(\text{LCBO-Ca}, x), \quad (6)$$

$$\Delta E_{\text{mutual energy}}(\text{LCBO-Ca}, x) = \Delta E_{\text{interface}}(\text{LCBO-Ca}, x) - x \cdot \Delta E_{\text{LCBO}} - (1-x) \cdot \Delta E_{\text{Ca}}, \quad (7)$$

where EP is the abbreviation of equilibrium phase, Ca is the abbreviation of cathode, $E(\text{EP})$ is determined by constructing the convex energy hull of all relevant phases in compositional phase diagram for the corresponding LCBO or cathode, $\Delta E_{\text{interface}}(\text{LCBO-Ca}, x)$ is the decomposition energy with respect to equilibrium phase of LCBO-Ca interface, $\Delta E_{\text{mutual energy}}(\text{LCBO-Ca}, x)$ the mutual energy between LCBO and cathode interface excluding the decomposition energy of LCBO and cathode themselves. The mutual reaction energies and equations are as a function of mixing ratio at this interface (x), and x varies from 0 to 1

For LCBO-Ca interfaces at the applied lithium potentials, the decomposition mutual reaction energy is defined in terms of $\Delta E_{D,\text{min},\text{mutual}}$ in equation (10) after considering the reaction equilibrium as well as the following equations:

$$\Delta E_{D,\text{interface}}(\text{LCBO-Ca}, x, \phi) = E(\text{EP}, x) - E_{\text{interface}}(\text{LCBO-Ca}, x) - \Delta n \cdot \mu_{\text{Li}}(\phi), \quad (8)$$

$$\Delta E_{D,\text{mutual}}(\text{LCBO-Ca}, x, \phi) = \Delta E_{D,\text{interface}}(\text{LCBO-Ca}, x, \phi) - x \cdot \Delta E_{\text{LCBO}}(\phi) - (1-x) \cdot \Delta E_{\text{Ca}}(\phi), \quad (9)$$

$$\Delta E_{D,\text{min},\text{mutual}} = \min(0 < x < 1) [\Delta E_{D,\text{mutual}}(\text{LCBO-Ca}, x, \phi)], \quad (10)$$

where $\Delta E_{D,\text{interface}}(\text{LCBO-Ca}, x, \phi)$ is the decomposition energy at applied potential, $\Delta E_{D,\text{mutual}}(\text{LCBO-Ca}, x, \phi)$ is the mutual energy between LCBO and cathode excluding the decomposition energy of LCBO and cathode themselves, $\Delta E_{D,\text{min},\text{mutual}}$ is minimum of the mutual reaction energy at the applied potential, and ϕ here varies from 2 to 5 V.

To search for the possible Li-ion migration pathways and the corresponding migration barriers, the AIMD and climbing image nudged elastic band (CI-NEB) methods are used.⁵⁷ Three initial images between two local energy minima structures are first set by linear interpolation and then fully relaxed. We estimate the diffusion coefficient via the following formula.

$$D_{\text{Li}} = \frac{1}{2} v (\Delta x)^2 e^{\left[-\frac{E_A}{k_B T}\right]}, \quad (11)$$

where Δx is the hop distance, v is the lattice vibrational frequency with a typical value of 10^{13} Hz, T is the temperature and k_B is the Boltzmann constant. E_A is the migration barrier. The Li-ion conductivity (σ_{Li}) is estimated from the Nernst-Einstein relation:

$$\sigma_{\text{Li}} = n_{\text{Li}} D_{\text{Li}} (q^2 / k_B T), \quad (12)$$

where n_{Li} is the concentration of mobile Li-ion, q is the charge of the diffusion carrier (Li), k_B is the Boltzmann constant, D_{Li} is calculated from equation (11), and T is temperature. In addition, the diffusion analyses are performed by using the pymatgen diffusion Python packages.⁴¹

3. RESULTS AND DISCUSSION

3.1 LCBO Structural Modeling

We use monoclinic Li_2CO_3 (C2/c structure) to model the LCBO system, which is composed of planar CO_3 units and LiO_4 tetrahedrons, as shown in Figures 1a and 1b. Planar CO_3 units form building blocks along the b -direction. Interconnected LiO_4 tetrahedrons along bc -plane are situated between these blocks. LiO_4 tetrahedrons are interconnected by corner- or edge-sharing. After B-doping, additional Li-ions are introduced into the framework because of the charge compensation, shown as Li interstitials in Figures 1c and 1d. These Li interstitials lead to a 3D connectivity of the LiO_4 tetrahedrons in LCBO. Here, the location of Li interstitials is suggested by previous calculation. Table 1 lists the calculation results of lattice parameters for LCBO system in comparison with that of Li_2CO_3 . The lattice constant of Li_2CO_3 is consistent with the experimental data (within 1% error). After B-doping, the lattice changes are within 5%, indicating B-dopants have only a minor effect on original structure. The density of states (DOS) of Li_2CO_3 and LCBO (in Figure S1) suggests that LCBO system has a bandgap as large as 4.4 eV. Therefore, LCBO is a good electron insulator, which is potent in blocking the electron leakage and preventing electrode corrosion.

3.2 Phase Stability and Electrochemical Stability of LCBO

To study the phase stability of LCBO, we use the grand canonical linear program (GCLP) and explore the phase diagram of Li_2O - B_2O_3 - CO_2 compounds with respect to a range of possible decomposed systems including binary and ternary, as shown in Figure 2a. For the decomposition of LCBO, the release of CO_2 gas is not associated with solid decomposition reaction. The only possible case is the

decomposition of quaternary LCBO ($0 < x < 0.625$) into ternary compounds, Li_2CO_3 and Li_3BO_3 , with decomposition energy from 17 to 47 meV/atom (Figure 2b). These values are much lower than that of $\text{Li}_{10}\text{GeP}_2\text{O}_{12}$ (56 meV/atom, decomposition into ternary Li_3PO_4 and Li_4GeO_4) and $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (69 meV/atom, decomposition into ternary Li_3PS_4 and Li_4GeS_4).⁴⁰ Experiments have demonstrated the synthesis of LGPO and LGPS and the success in battery setup test.⁵⁸⁻⁵⁹ The reason for the stability performance of them is due to the very high kinetic barrier, which prohibits the solid-solid phase decomposition. Thus, the solid decomposition is less a problem in practice because of the kinetic obstruction. In addition, higher B-dopant content in LCBO leads to higher Li interstitial content, but the worse stability (high decomposition energy). To balance them, we choose LCBO ($x = 0.375$) to be the optimized structure, at which point the decomposition energy is relatively close to zero while ensuring Li interstitial content, as shown in Figure 2b.

Achieving a broad electrochemical window in ISEs is very challenging in the battery system with a high-voltage cathode and a low-voltage anode because of the extreme conditions of Li chemical potential and chemical reaction between ISEs and electrodes.^{4, 60} Here, the (electro)chemical stability of the LCBO system is divided into two stages: electrochemical stability and chemical stability. For electrochemical stability, using the grand potential phase diagram, we calculate the equilibrium of LCBO against a range of Li chemical potential (μ_{Li}) or voltage (ϕ) to show the electrochemical stability and the voltage range where the material is stable. Figure 3 shows the electrochemical window (upper inset), phase equilibria (middle inset), and Li update versus voltage (lower inset) for LCBO system with Li_2CO_3 and Li_3BO_3 as references. The detailed phase equilibrium at reduction and oxidation potentials of LCBO are listed in Table S1. We find that both Li_2CO_3 and Li_3BO_3 has a

wide electrochemical window from a reduction potential of 1.27 and 0.30 V to an oxidation potential of 4.11 and 3.47 V (green bars in upper inset of Figure 3). For LCBO, when an applied potential range is from 1.27 to 3.47 V (yellow bar in upper inset of Figure 3), LCBO decomposes into Li_2CO_3 and Li_3BO_3 due to the voltage-induced thermodynamic driving force. However, it is noticeable that this voltage range (from 1.27 to 3.47 V) is right in the stability window of Li_2CO_3 and Li_3BO_3 . The SEI-like materials can suppress the further decomposition of LCBO since Li_2CO_3 and Li_3BO_3 can block electrons and allow Li-ion transport. Therefore, the stable voltage range for LCBO is from 1.27 to 3.47 V. Recent experimental synthesis and electrochemical characterization of Li_2CO_3 - Li_3BO_3 has demonstrated its phase stability and high ionic conductivity.²⁹ Thus, the voltage-induced instability is less problematic in practice due to the good electronic insulation and ionic conduction of Li_2CO_3 - Li_3BO_3 .

It is important to know whether the ISE-electrode interface reacts exothermically or whether the interface is chemical stability, which could be occurring in heat treatment of cell preparation or (dis)charging of cell testing. We therefore extend the above methodology to calculate the mutual reaction energy of $\Delta E_{\text{mutual energy}}$ and $\Delta E_{\text{D,min,mutual}}$ (see eq. (3-10) in the methods part), as shown in upper and lower insets of Figure 4, respectively. The cathodes, LiXO_2 ($\text{X} = \text{Co}, \text{Mn}, \text{and Ni}$), LiFePO_4 , and LiMn_2O_4 , are chosen as electrodes. The phase equilibria of ISE-electrode interfaces are listed in Table S2 and S3 of supporting information. In the absence of any applied chemical potential, these ISEs show good interfacial stability with decomposition energy from -57 to 0 meV/atom. In contrast, the decomposition energy between Li_3P_4 and LiCoO_2 is about -405 meV/atom. The interfaces of Li_2CO_3 with all cathodes are highly stable, and LCBO is non-reactive with LiXO_2 . On the other hand,

applying Li chemical potential over the interfaces, the interfacial stability of ISEs is weakened with some cathodes, especially with LiNiO₂. The $\Delta E_{D,min,muual}$ energy of LCBO with cathodes except LiFePO₄ and LiNiO₂ is only in the -50 to 0 meV/atom range compared to that of garnet Li₇La₃Zr₂O₁₂ (LLZO) with LiCoO₂ cathode (from -33 to 0 meV/atom).⁵⁵ The good stability of Li₂CO₃-Li₃BO₃ and LLZO with LiCoO₂ cathode has been observed in the experiments.²⁹ The interfacial decomposition reactions may be kinetically inhibited. Thus, we suggest that LiCoO₂, LiMnO₂, and LiMn₂O₄ are suitable cathodes for LCBO, which is very helpful to improve high-voltage performance of ASSLBs. In addition, for all $\Delta E_{D,min,muual} < 0$ interfacial reactions, the phase Li-metal is in the final phase equilibria, which may lead to formation of thick interphase layers or potential degradation at the interface. Hence, further studies are awaited to evaluate their characteristics.

3.3 Li-ion Transport in LCBO (x = 0.375)

As suggested by Shi *et al.*,⁶¹⁻⁶² Li-ion conductivity in Li₂CO₃ stems from the migration of intrinsic point defect (i.e. Li interstitial), and occurs through a knock-off or cooperative mechanism involving the concerted knock-off motion of interstitial and lattice lithium.⁶³⁻⁶⁴ Such a cooperative migration mechanism has been elucidated in some ISEs.^{52, 65-66} Here, the interstitial site belongs to a LiO₄ tetrahedron and connects two parallel migration paths (along *bc*-plane). Then, there is a three-dimensional (3D) migration channel in LCBO.

Next, we investigate the migration of Li interstitial in LCBO. The energy profile, schematic pathways and transition state for Li cooperative migration are shown in Figure 5. In the whole migration coordination, Li interstitial will push the neighboring lattice Li into the target interstitial site while the lattice site is filled with the original Li interstitial. The overall diffusion barrier for the Li-ion

interstitial in LCBO is about 0.23 ± 0.02 eV, the average migration distance for each step is about 2.60 Å. For LCBO, the concentration of mobile Li-ions is $\sim 5.54 \times 10^{21} \text{ cm}^{-3}$. Thus, according to the equations (11-12), the calculated diffusion coefficient of D_{Li} is about $6.0 \times 10^{-7} \text{ cm}^2 \text{ s}^{-1}$ at room temperature (300 K). Moreover, the calculated Li-ionic conductivity of σ_{Li} is about $17 \pm 10 \text{ mS cm}^{-1}$ under room temperature. Ionic trajectories are analyzed to visualize the migration paths in Figure 6. This figure shows that Li-ion migration within this system is slightly anisotropic. Most diffusion is restricted to within the LiO_4 tetrahedra layers along bc -plane with small evidence of Li-ion diffusion between layers, suggesting a quasi-isotropic three-dimensional diffusion channel. Obviously, the ionic conductivity of LCBO is superior to most of current ISE materials, such as LLZO¹⁷ and $\text{Li}_{1+x}\text{Al}_x\text{Ti}_{1-x}(\text{PO}_4)_3$,⁶⁷. Thus, LCBO is a potential ISE material in next-generation Li battery.

4. CONCLUSIONS

In summary, by a combination of atomistic simulation and materials project database, we investigate the electrochemical properties of LCBO as a novel solid-state electrolyte. It is shown that LCBO is a good electronic insulator with a large bandgap of 4.3 eV and B-doping induces slight changes on lattice parameters. Our results suggest that the stability of LCBO phase and interface with electrodes are dependent on the applied voltage. The electrochemical stability studies suggest that LCBO will decompose into Li_2CO_3 and Li_3BO_3 under the voltage from 1.27 to 3.46 V. The formed phase equilibria (Li_2CO_3 and Li_3BO_3) has good electrochemical window and good chemical stability with layered LiCoO_2 and spinel LiMn_2O_4 cathodes. Meanwhile, the corresponding mutual reaction energies of LCBO with LiCoO_2 , LiMnO_2 , and LiMn_2O_4 are only in the range of -50 to 0 meV/atom,

suggesting a small thermodynamic driving force. We suggest LCBO has very good chemical stability in contact with these cathodes. Unfortunately, phase Li-metal exists in the formed phase equilibria between LCBO and cathodes, which leads to electron leakage and electrolyte corrosion. Therefore, further experimental examination is very necessary. The Li-ion cooperative migration has the migration barrier as low as 0.23 ± 0.02 eV and possesses high ionic conductivity in a quasi-isotropic three-dimensional diffusion channel. Overall, LCBO is a potential candidate for ISE materials in ASSLBs.

ASSOCIATED CONTENT

Supporting Information

Supplementary information is available free of charge on ACS Publications Website.

Total density of states for Li_2CO_3 and LCBO ($x = 0.375$), the phase equilibria at the reduction and oxidation potentials of LCBO, electrolyte-electrode interfaces.

Author Information

(Bingkai Zhang) *E-mail: zhangbk@gdut.edu.cn

(Zhan Lin) *E-mail: zhanlin@gdut.edu.cn

(Feng Pan) *E-mail: panfeng@pku.edu.cn

Competing Financial Interests

The authors declare no competing financial interests.

ACKNOWLEDGMENTS

This work was supported by This work was financially supported by startup R&D funding from the

One-Hundred Young Talents Program of Guangdong University of Technology (No. 22041331901), National Key R&D Program of China (2016YFB0700600), Soft Science Research Project of Guangdong Province (No. 2017B030301013), Shenzhen Science and Technology Research Grant (ZDSYS201707281026184), National Natural Science Foundation of China (No. 51703070). Wang was supported by the Assistant Secretary for Energy Efficiency and Renewal Energy of the U.S. Department of Energy under the Battery Materials Research (BMR) program.

REFERENCE

- (1) Armand, M.; Tarascon, J. M. Building Better Batteries. *Nature* **2008**, *451*, 652-657.
- (2) Janek, J.; Zeier, W. G. A Solid Future for Battery Development. *Nat. Energy* **2016**, *1*, 16141.
- (3) Manthiram, A.; Yu, X.; Wang, S. Lithium Battery Chemistries Enabled by Solid-State Electrolytes. *Nat. Rev. Mater.* **2017**, *2*, 16103.
- (4) Famprikis, T.; Canepa, P.; Dawson, J. A.; Islam, M. S.; Masquelier, C. Fundamentals of Inorganic Solid-State Electrolytes for Batteries. *Nat. Mater.* **2019**, *18*, 1278-1291.
- (5) Bachman, J. C.; Muy, S.; Grimaud, A.; Chang, H.-H.; Pour, N.; Lux, S. F.; Paschos, O.; Maglia, F.; Lupart, S.; Lamp, P.; Giordano, L.; Shao-Horn, Y. Inorganic Solid-State Electrolytes for Lithium Batteries: Mechanisms and Properties Governing Ion Conduction. *Chem. Rev.* **2016**, *116*, 140-162.
- (6) Yu, C.; Ganapathy, S.; Eck, E. R. H. v.; Wang, H.; Basak, S.; Li, Z.; Wagemaker, M. Accessing the Bottleneck in All-Solid State Batteries, Lithium-ion Transport over the Solid-Electrolyte-Electrode Interface. *Nat. Commun.* **2017**, *8*, 1086.
- (7) Zhang, B.; Tan, R.; Yang, L.; Zheng, J.; Zhang, K.; Mo, S.; Lin, Z.; Pan, F. Mechanisms and Properties of Ion-transport in Inorganic Solid Electrolytes. *Energy Stor. Mater.* **2018**, *10*, 139-159.
- (8) Kato, Y.; Hori, S.; Saito, T.; Suzuki, K.; Hirayama, M.; Mitsui, A.; Yonemura, M.; Iba, H.; Kanno,

- R. High-Power All-Solid-State Batteries using Sulfide Superionic Conductors. *Nat. Energy* **2016**, *1*, 16030.
- (9) Liang, J.; Li, X.; Zhao, Y.; Goncharova, L. V.; Wang, G.; Adair, K. R.; Wang, C.; Li, R.; Zhu, Y.; Qian, Y.; Zhang, L.; Yang, R.; Lu, S.; Sun, X. In Situ Li_3PS_4 Solid-State Electrolyte Protection Layers for Superior Long-Life and High-Rate Lithium-Metal Anodes. *Adv. Mater.* **2018**, *30*, 1804684.
- (10) Liu, Z.; Fu, W.; Payzant, E. A.; Yu, X.; Wu, Z.; Dudney, N. J.; Kiggans, J.; Hong, K.; Rondinone, A. J.; Liang, C. Anomalous High Ionic Conductivity of Nanoporous $\beta\text{-Li}_3\text{PS}_4$. *J. Am. Chem. Soc.* **2013**, *135*, 975-978.
- (11) Lang, B.; Ziebarth, B.; Elsässer, C. Lithium Ion Conduction in $\text{LiTi}_2(\text{PO}_4)_3$ and Related Compounds Based on the NASICON Structure: A First-Principles Study. *Chem. Mater.* **2015**, *27*, 5040-5048.
- (12) Kahlaoui, R.; Arbi, K.; Sobrados, I.; Jimenez, R.; Sanz, J.; Ternane, R. Cation Miscibility and Lithium Mobility in NASICON $\text{Li}_{1+x}\text{Ti}_{2-x}\text{Sc}_x(\text{PO}_4)_3$ ($0 \leq x \leq 0.5$) Series: A Combined NMR and Impedance Study. *Inorg. Chem.* **2017**, *56*, 1216-1224.
- (13) Bernges, T.; Culver, S. P.; Minafra, N.; Koerver, R.; Zeier, W. G. Competing Structural Influences in the Li Superionic Conducting Argyrodites $\text{Li}_6\text{PS}_{5-x}\text{Se}_x\text{Br}$ ($0 \leq x \leq 1$) upon Se Substitution. *Inorg. Chem.* **2018**, *57*, 13920-13928.
- (14) Yu, C.; Ganapathy, S.; de Klerk, N. J.; Roslon, I.; van Eck, E. R.; Kentgens, A. P.; Wagemaker, M. Unravelling Li-ion Transport from Pico-seconds to Seconds: Bulk Versus Interfaces in an Argyrodite $\text{Li}_6\text{PS}_5\text{Cl-Li}_2\text{S}$ All Solid State Li-ion Battery. *J. Am. Chem. Soc.* **2016**, *138*, 11192-11201.
- (15) Deiseroth, H. J.; Kong, S. T.; Eckert, H.; Vannahme, J.; Reiner, C.; Zaiss, T.; Schlosser, M. $\text{Li}_6\text{PS}_5\text{X}$: A Class of Crystalline Li-rich Solids with an Unusually High Li^+ Mobility. *Angew. Chem.*

Int. Ed. **2008**, *47*, 755-758.

(16) Jalem, R.; Rushton, M. J. D.; Manalastas, W.; Nakayama, M.; Kasuga, T.; Kilner, J. A.; Grimes, R. W. Effects of Gallium Doping in Garnet-Type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ Solid Electrolytes. *Chem. Mater.* **2015**, *27*, 2821-2831.

(17) Chen, Y.; Rangasamy, E.; Liang, C.; An, K. Origin of High Li^+ Conduction in Doped $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ Garnets. *Chem. Mater.* **2015**, *27*, 5491-5494.

(18) Zhao, Y.; Daemen, L. L. Superionic Conductivity in Lithium-Rich Anti-Perovskites. *J. Am. Chem. Soc.* **2012**, *134*, 15042-15047.

(19) Deng, Z.; Radhakrishnan, B.; Ong, S. P. Rational Composition Optimization of the Lithium-Rich $\text{Li}_3\text{OCl}_{1-x}\text{Br}_x$ Anti-Perovskite Superionic Conductors. *Chem. Mater.* **2015**, *27*, 3749-3755.

(20) Tian, Y.; Shi, T.; Richards, W. D.; Li, J.; Kim, J. C.; Bo, S.-H.; Ceder, G. Compatibility Issues between Electrodes and Electrolytes in Solid-State Batteries. *Energy Environ. Sci.* **2017**, *10*, 1150-1166.

(21) Tian, Y.; Sun, Y.; Hannah, D. C.; Xiao, Y.; Liu, H.; Chapman, K. W.; Bo, S.-H.; Ceder, G. Reactivity-Guided Interface Design in Na Metal Solid-State Batteries. *Joule* **2019**, *3*, 1037-1050.

(22) Pan, J.; Zhang, Q.; Xiao, X.; Cheng, Y.-T.; Qi, Y. Design of Nanostructured Heterogeneous Solid Ionic Coatings through a Multiscale Defect Model. *ACS Appl. Mater. Interfaces* **2016**, *8*, 5687-5693.

(23) Kim, A. Y.; Strauss, F.; Bartsch, T.; Teo, J. H.; Hatsukade, T.; Mazilkin, A.; Janek, J.; Hartmann, P.; Brezesinski, T. Stabilizing Effect of a Hybrid Surface Coating on a Ni-Rich NCM Cathode Material in All-Solid-State Batteries. *Chem. Mater.* **2019**, *31*, 9664-9672.

(24) Shi, S.; Qi, Y.; Li, H.; Hector, L. G. Defect Thermodynamics and Diffusion Mechanisms in

Li₂CO₃ and Implications for the Solid Electrolyte Interphase in Li-Ion Batteries. *J. Phys. Chem. C* **2013**, *117*, 8579-8593.

(25) Shi, S.; Lu, P.; Liu, Z.; Qi, Y.; Hector, L. G.; Li, H.; Harris, S. J. Direct Calculation of Li-Ion Transport in the Solid Electrolyte Interphase. *J. Am. Chem. Soc.* **2012**, *134*, 15476-15487.

(26) Li, Y.; Qi, Y. Energy Landscape of the Charge Transfer Reaction at the Complex Li/SEI/Electrolyte Interface. *Energy Environ. Sci.* **2019**, *12*, 1286-1295.

(27) Dissanayake, M. A. K. L.; Mellander, B. E. Phase Diagram and Electrical Conductivity of the Li₂SO₄-Li₂CO₃ System. *Solid State Ionics* **1986**, *21*, 279-285.

(28) Shannon, R. D.; Taylor, B. E.; English, A. D.; Berzins, T. New Li Solid Electrolytes. *Electrochim. Acta* **1977**, *22*, 783-796.

(29) Jung, S. H.; Oh, K.; Nam, Y. J.; Oh, D. Y.; Br uner, P.; Kang, K.; Jung, Y. S. Li₃BO₃-Li₂CO₃: Rationally Designed Buffering Phase for Sulfide All-Solid-State Li-Ion Batteries. *Chem. Mater.* **2018**, *30*, 8190-8200.

(30) Richards, W. D.; Wang, Y.; Miara, L. J.; Kim, J. C.; Ceder, G. Design of Li_{1+2x}Zn_{1-x}PS₄, a New Lithium Ion Conductor. *Energy Environ. Sci.* **2016**, *9*, 3272-3278.

(31) Wang, Y.; Richards, W. D.; Ong, S. P.; Miara, L. J.; Kim, J. C.; Mo, Y.; Ceder, G. Design Principles for Solid-State Lithium Superionic Conductors. *Nat. Mater.* **2015**, *14*, 1026-1031.

(32) Liu, G.; Xie, D.; Wang, X.; Yao, X.; Chen, S.; Xiao, R.; Li, H.; Xu, X. High Air-Stability and Superior Lithium Ion Conduction of Li_{3+3x}P_{1-x}Zn_xS_{4-x}O_x by Aliovalent Substitution of ZnO for All-Solid-State Lithium Batteries. *Energy Stor. Mater.* **2019**, *17*, 266-274.

(33) Lei, X.; Wu, W.; Xu, B.; Ouyang, C.; Huang, K. LiGaOS is a Fast Li-Ion Conductor: A First-Principles Prediction. *Mater. Des.* **2019**, 108264.

- (34) Yang, Y.; Wu, Q.; Cui, Y.; Chen, Y.; Shi, S.; Wang, R.-Z.; Yan, H. Elastic Properties, Defect Thermodynamics, Electrochemical Window, Phase Stability, and Li⁺ Mobility of Li₃PS₄: Insights from First-Principles Calculations. *ACS Appl. Mater. Interfaces* **2016**, *8*, 25229-25242.
- (35) Zhang, B.; Yang, L.; Wang, L.-W.; Pan, F. Cooperative Transport Enabling Fast Li-ion Diffusion in Thio-LISICON Li₁₀SiP₂S₁₂ Solid Electrolyte. *Nano Energy* **2019**, *62*, 844-852.
- (36) Mo, Y.; Ong, S. P.; Ceder, G. First Principles Study of the Li₁₀GeP₂S₁₂ Lithium Super Ionic Conductor Material. *Chem. Mater.* **2012**, *24*, 15-17.
- (37) Ong, S. P.; Mo, Y.; Richards, W. D.; Miara, L.; Lee, H. S.; Ceder, G. Phase Stability, Electrochemical Stability and Ionic Conductivity of the Li_{10±1}MP₂X₁₂ (M = Ge, Si, Sn, Al or P, and X = O, S or Se) Family of Superionic Conductors. *Energy Environ. Sci.* **2012**, *6*, 148-156.
- (38) Deng, Y.; Eames, C.; Chotard, J.-N.; Lalère, F.; Seznec, V.; Emge, S.; Pecher, O.; Grey, C. P.; Masquelier, C.; Islam, M. S. Structural and Mechanistic Insights into Fast Lithium-Ion Conduction in Li₄SiO₄-Li₃PO₄ Solid Electrolytes. *J. Am. Chem. Soc.* **2015**, *137*, 9136-9145.
- (39) He, X.; Zhu, Y.; Epstein, A.; Mo, Y. Statistical Variances of Diffusional Properties from *ab initio* Molecular Dynamics Simulations. *NPJ Comput. Mater.* **2018**, *4*, 18.
- (40) Ong, S. P.; Mo, Y.; Richards, W. D.; Miara, L.; Lee, H. S.; Ceder, G. Phase stability, Electrochemical Stability and Ionic Conductivity of the Li_{10+/-1}MP₂X₁₂ (M = Ge, Si, Sn, Al or P, and X = O, S or Se) Family of Superionic Conductors. *Energy Environ. Sci.* **2013**, *6*, 148-156.
- (41) Deng, Z.; Zhu, Z.; Chu, I.-H.; Ong, S. P. Data-Driven First-Principles Methods for the Study and Design of Alkali Superionic Conductors. *Chem. Mater.* **2017**, *29*, 281-288.
- (42) Ong, S. P.; Richards, W. D.; Jain, A.; Hautier, G.; Kocher, M.; Cholia, S.; Gunter, D.; Chevrier, V. L.; Persson, K. A.; Ceder, G. Python Materials Genomics (pymatgen): A Robust, Open-Source

- Python Library for Materials Analysis. *Comput. Mater. Sci.* **2013**, *68*, 314-319.
- (43) Ong, S. P.; Jain, A.; Hautier, G.; Kang, B.; Ceder, G. Thermal Stabilities of Delithiated Olivine MPO_4 (M=Fe, Mn) Cathodes Investigated using First Principles Calculations. *Electrochem. Commun.* **2010**, *12*, 427-430.
- (44) Ong, S. P.; Wang, L.; Kang, B.; Ceder, G. Li-Fe-P-O₂ Phase Diagram from First Principles Calculations. *Chem. Mater.* **2008**, *20*, 1798-1807.
- (45) Xiao, Y.; Wang, Y.; Bo, S.-H.; Kim, J. C.; Miara, L. J.; Ceder, G. Understanding Interface Stability in Solid-State Batteries. *Nat. Rev. Mater.* **2019**. doi:10.1038/s41578-019-0157-5.
- (46) Kresse, G.; Furthmüller, J. Efficient Iterative Schemes for *ab initio* Total-energy Calculations using a Plane-wave Basis Set. *Phys. Rev. B* **1996**, *54*, 11169-11186.
- (47) Blöchl, P. E. Projector Augmented-wave Method. *Phys. Rev. B* **1994**, *50*, 17953-17979.
- (48) Perdew, J. P.; Burke, K.; Ernzerhof, M. Generalized Gradient Approximation Made Simple. *Phys. Rev. Lett.* **1996**, *77*, 3865-3868.
- (49) Hasnip, P. J.; Refson, K.; Probert, M. I. J.; Yates, J. R.; Clark, S. J.; Pickard, C. J. Density Functional Theory in the Solid State. *Philos. Trans. R. Soc. A* **2014**, *372*, 20130270.
- (50) Meng, Y. S.; Arroyo-de Dompablo, M. E. First Principles Computational Materials Design for Energy Storage Materials in Lithium Ion Batteries. *Energy Environ. Sci.* **2009**, *2*, 589-609.
- (51) Kirklin, S.; Saal, J. E.; Meredig, B.; Thompson, A.; Doak, J. W.; Aykol, M.; Rühl, S.; Wolverton, C. The Open Quantum Materials Database (OQMD): Assessing the Accuracy of DFT Formation Energies. *NPJ Comput. Mater.* **2015**, *1*, 15010.
- (52) Monchak, M.; Hupfer, T.; Senyshyn, A.; Boysen, H.; Chernyshov, D.; Hansen, T.; Schell, K. G.; Bucharsky, E. C.; Hoffmann, M. J.; Ehrenberg, H. Lithium Diffusion Pathway in $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$

(LATP) Superionic Conductor. *Inorg. Chem.* **2016**, *55*, 2941-2945.

(53) Nolan, A. M.; Zhu, Y.; He, X.; Bai, Q.; Mo, Y. Computation-Accelerated Design of Materials and Interfaces for All-Solid-State Lithium-Ion Batteries. *Joule* **2018**, *2*, 2016-2046.

(54) Han, F.; Zhu, Y.; He, X.; Mo, Y.; Wang, C. Electrochemical Stability of $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ and $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ Solid Electrolytes. *Adv. Energy Mater.* **2016**, *6*, 1501590.

(55) Zhu, Y.; He, X.; Mo, Y. First Principles Study of Electrochemical and Chemical Stability of the Solid Electrolyte-Electrode Interfaces in All-Solid-State Li-Ion Batteries. *J. Mater. Chem. A* **2015**, *4*, 3253-3266.

(56) Zhu, Y.; He, X.; Mo, Y. Origin of Outstanding Stability in the Lithium Solid Electrolyte Materials: Insights from Thermodynamic Analyses Based on First-Principles Calculations. *ACS Appl. Mater. Interfaces* **2015**, *7*, 23685-23693.

(57) Henkelman, G.; Uberuaga, B. P.; Jónsson, H. A Climbing Image Nudged Elastic Band Method for Finding Saddle Points and Minimum Energy Paths. *J. Chem. Phys.* **2000**, *113*, 9901-9904.

(58) Song, S.; Dong, Z.; Deng, F.; Hu, N. Lithium Superionic Conductors $\text{Li}_{10}\text{MP}_2\text{O}_{12}$ (M = Ge, Si). *Functional Materials Letters* **2018**, *11*, 1850039.

(59) Kamaya, N.; Homma, K.; Yamakawa, Y.; Hirayama, M.; Kanno, R.; Yonemura, M.; Kamiyama, T.; Kato, Y.; Hama, S.; Kawamoto, K.; Mitsui, A. A Lithium Superionic Conductor. *Nat. Mater.* **2011**, *10*, 682-686.

(60) Fitzhugh, W.; Wu, F.; Ye, L.; Deng, W.; Qi, P.; Li, X. A High-Throughput Search for Functionally Stable Interfaces in Sulfide Solid-State Lithium Ion Conductors. *Adv. Energy Mater.* **2019**, *9*, 1900807.

(61) Shi, S.; Lu, P.; Liu, Z.; Qi, Y.; Hector Jr, L. G.; Li, H.; Harris, S. J. Direct Calculation of Li-ion

Transport in the Solid Electrolyte Interphase. *J. Am. Chem. Soc.* **2012**, *134*, 15476-15487.

(62) Shi, S.; Qi, Y.; Li, H.; Hector Jr, L. G. Defect Thermodynamics and Diffusion Mechanisms in Li_2CO_3 and Implications for the Solid Electrolyte Interphase in Li-ion Batteries. *J. Phys. Chem. C* **2013**, *117*, 8579-8593.

(63) He, X.; Zhu, Y.; Mo, Y. Origin of Fast Ion Diffusion in Super-ionic Conductors. *Nat. Commun.* **2017**, *8*, 15893.

(64) Zhang, B.; Yang, L.; Wang, L.-W.; Pan, F. Cooperative Transport Enabling Fast Li-ion Diffusion in Thio-LISICON $\text{Li}_{10}\text{SiP}_2\text{S}_{12}$ Solid Electrolyte. *Nano Energy* **2019**, *62*, 844-852.

(65) Zhang, Z.; Zou, Z.; Kaup, K.; Xiao, R.; Shi, S.; Avdeev, M.; Hu, Y. S.; Wang, D.; He, B.; Li, H. Correlated Migration Invokes Higher Na^+ -Ion Conductivity in NaSICON-Type Solid Electrolytes. *Adv. Energy Mater.* **2019**, *9*, 1902373.

(66) Zhang, B.; Lin, Z.; Dong, H.; Wang, L.-W.; Pan, F. Revealing Cooperative Li-ion Migration in $\text{Li}_{1+x}\text{Al}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ Solid State Electrolytes with High Al Doping. *J. Mater. Chem. A* **2020**, *8*, 342-348.

(67) Deng, Z.; Mo, Y.; Ong, S. P. Computational Studies of Solid-state Alkali Conduction in Rechargeable Alkali-ion Batteries. *NPG Asia Mater.* **2016**, *8*, e254.

(68) Idemoto, Y.; Richardson, J. W.; Koura, N.; Kohara, S.; Loong, C.-K. Crystal Structure of $(\text{Li}_x\text{K}_{1-x})_2\text{CO}_3$ ($x = 0, 0.43, 0.5, 0.62, 1$) by Neutron Powder Diffraction Analysis. *J. Phys. Chem. Solids* **1998**, *59*, 363-376.

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

Table 1. Experimental and calculated structural parameters for Li₂CO₃ and LCBO crystal (space group C2/c).

	a (Å)	b (Å)	c (Å)	α (°)	β (°)	γ (°)
Li ₂ CO ₃	8.387	5.021	6.318	90.00	114.967	90.00
Li ₂ CO ₃ -Exp. ⁶⁸	8.359	4.974	6.194	90.00	114.789	90.00
Li _{2.125} C _{0.875} B _{0.125} O ₃	8.412	4.982	6.413	90.523	113.193	89.732
Li _{2.25} C _{0.75} B _{0.25} O ₃	8.468	4.986	6.501	89.821	113.012	90.132
Li _{2.375} C _{0.625} B _{0.375} O ₃	8.443	4.978	6.603	89.556	113.697	89.753
Li _{2.5} C _{0.5} B _{0.5} O ₃	8.404	5.013	6.793	89.945	114.073	89.687
Li _{2.625} C _{0.375} B _{0.625} O ₃	8.381	5.012	6.696	88.267	113.543	90.263

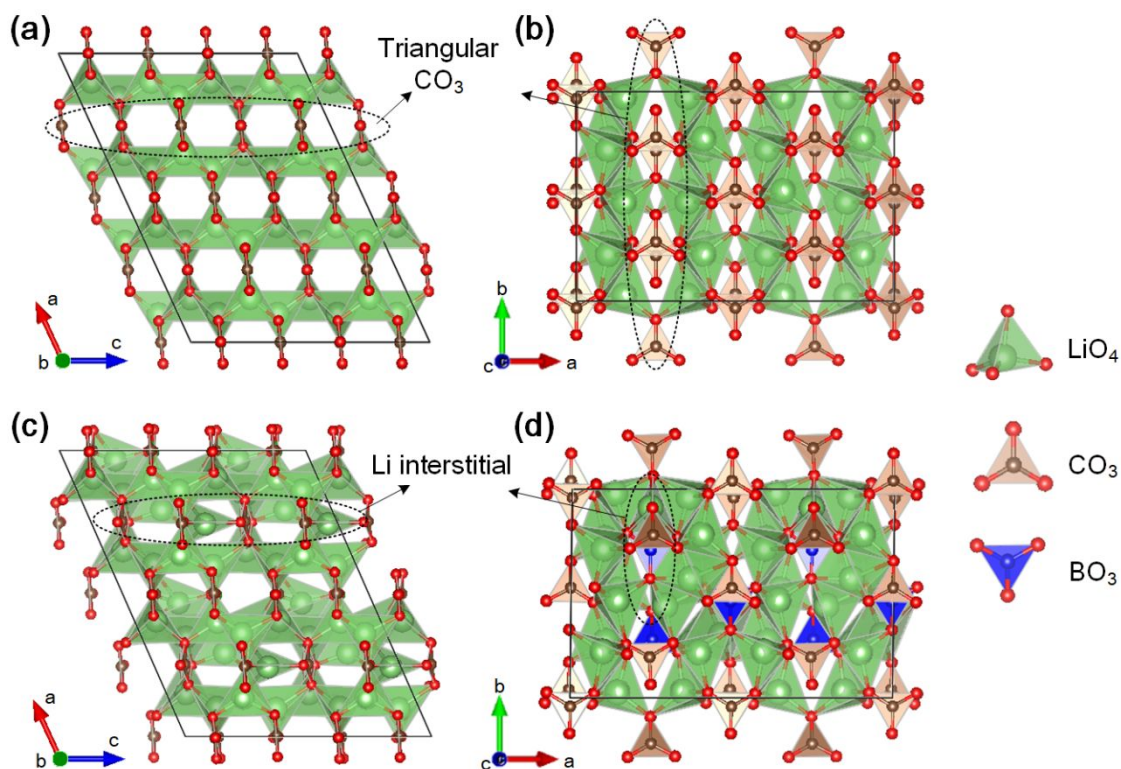


Figure 1. Polyhedral view of the Li_2CO_3 and LCBO crystal structure projected along the b-axis (a, c) and c-axis (b, d). Dotted lines highlight triangular CO_3 blocks and Li interstitial. Tetrahedral LiO_4 , triangular CO_3 and BO_3 units shown in green, light brown and light blue, respectively.

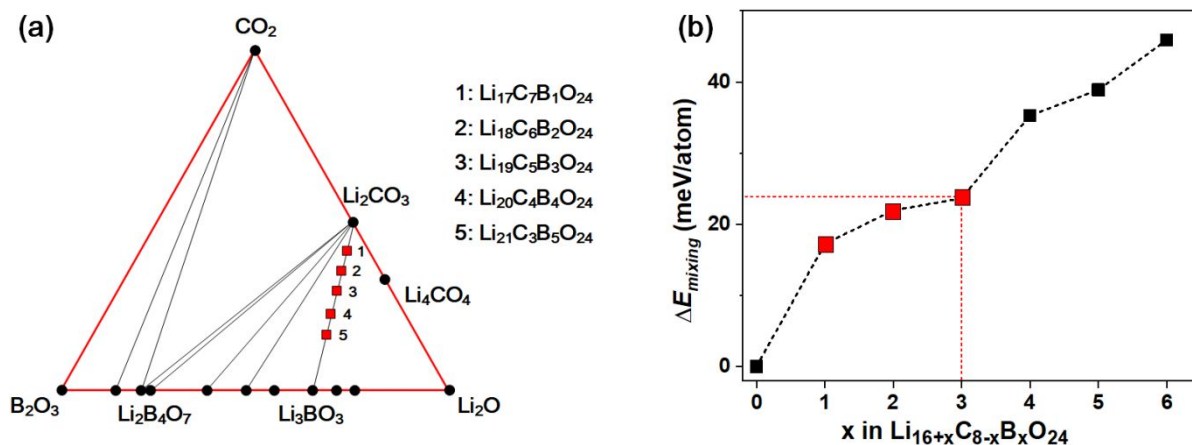


Figure 2. (a) Li-C-B-O ternary phase diagram assessing the computed stability of compounds formed by B-doping in $\text{Li}_{16+x}\text{C}_{8-x}\text{B}_x\text{O}_{24}$. Results showed in the diagram starting from stoichiometric Li_2O , Ba_2O_3 , and CO_2 . Black solid circles indicate clearly stable phases. Red solid squares indicate metastable phases. (b) The mixing energy in $\text{Li}_{16+x}\text{C}_{8-x}\text{B}_x\text{O}_{24}$. Cases where the mixing energy was close to zero, are shown with red solid squares.

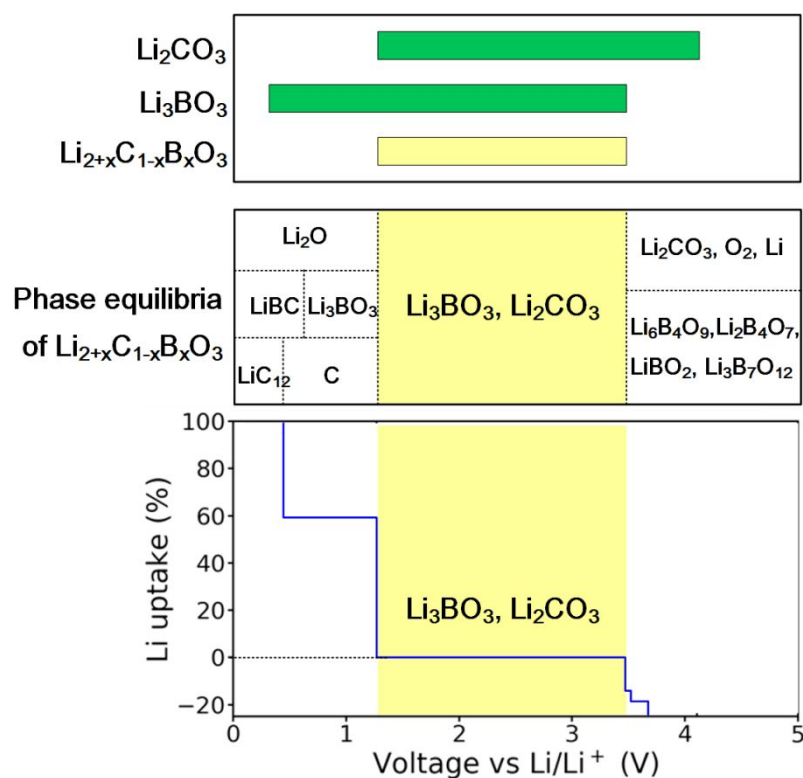


Figure 3. Electrochemical stability of studied solid electrolytes (top). Predicted phase equilibria over different voltage (μ_{Li}) range for LCBO (middle). Li uptake per formula unit of LCBO vs voltage (bottom). Green bars indicate stable electrochemical windows, and yellow bars indicate extended electrochemical windows.

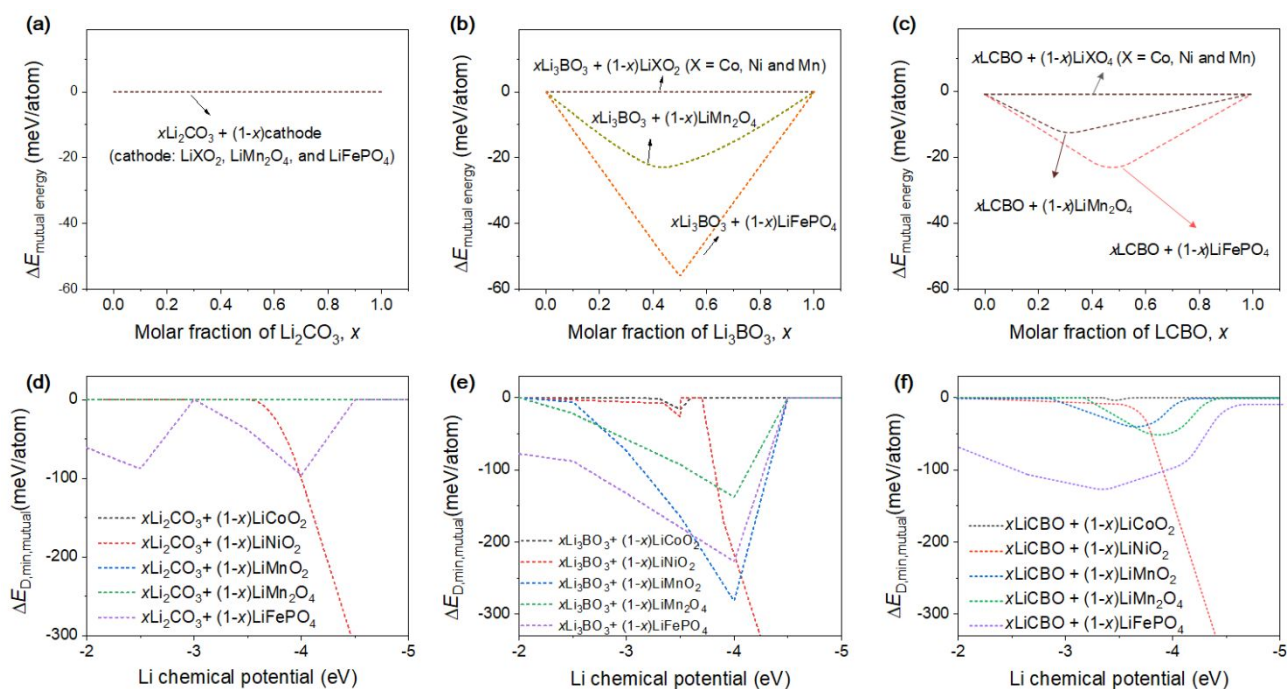


Figure 4. (a), (b), and (c) The calculated mutual reaction energy ΔE_{mutual} of Li_2CO_3 , Li_3BO_3 , and LCBO ($x = 0.375$) in contact with different cathodes without any applied chemical potential. The chemical stability of interfaces in the three figures is about whether two materials would react exothermically without any applied chemical potential. (d), (e), and (f) The calculated minimum mutual reaction energy $\Delta E_{\text{D,min,mutual}}$ of Li_2CO_3 , Li_3BO_3 , and LCBO in contact with different cathodes, in which interfaces open to different Li chemical potential. The cathode is one of LiXO_2 (X = Co, Mn and Ni), LiFePO_4 , and LiMn_2O_4 . Some ΔE_{mutual} and $\Delta E_{\text{D,min,mutual}}$ energies have zero or near-zero values, which overlap at the $\Delta E = 0$. A negative value of ΔE suggests the thermodynamically favorable reaction of interface. Therefore, the $\Delta E_{\text{D,min,mutual}}$ energy here is corresponding to the most favorable decomposition energy at a ratio of x.

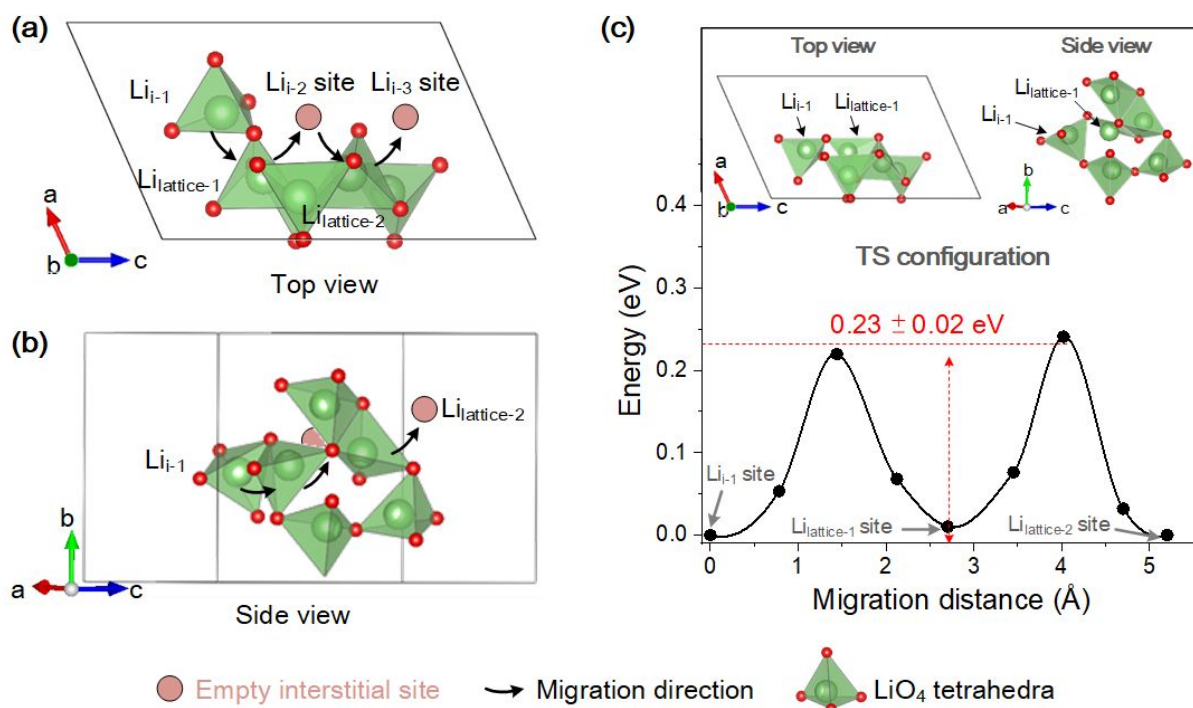


Figure 5. (a) Top view and (b) side view for interconnecting LiO₄ tetrahedra arranged along interstitial Li diffusion pathway. (c) The corresponding migration barriers (lower inset) and transition state (TS) configuration (upper inset). Here, Li_{i-1} and Li_{i-2} are two interstitial sites, and Li_{lattice-1} and Li_{lattice-2} are two lattice sites. In the TS configuration, Li_{i-1} “attacks” the oxygen atoms of lattice LiO₄ tetrahedra, forming bonds with them, and leading to the lattice Li-ion “departing” the same lattice LiO₄ tetrahedra.

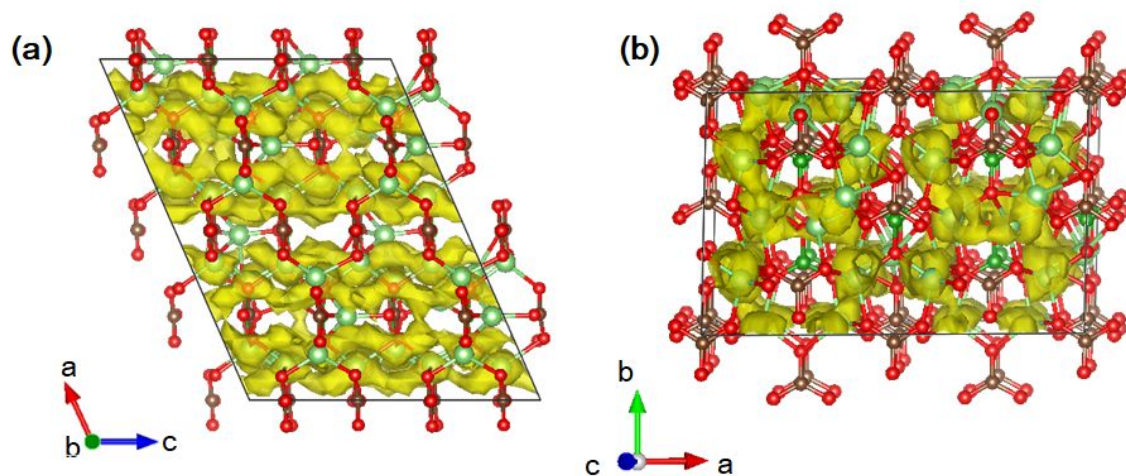


Figure 6. Iso-surfaces (shown by yellow) of the ionic probability densities evaluated from ionic trajectories calculated over 60 ps AIMD at 800 K. (a) Top view and (b) stereo view of LCBO structure.

TOC

